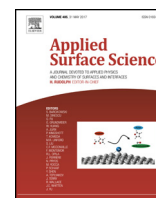




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Full length article

Effect of irradiation on the surface morphology of nanostructured superhydrophobic surfaces fabricated by ion beam irradiation

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ABSTRACT

Superhydrophobic surfaces exhibit anti-smudging, anti-sticking, anti-icing, and self-cleaning properties. Superhydrophobic surfaces with nanostructured morphologies were fabricated using an ion beam irradiation method. The ion beam irradiation method was employed to gain insight into the irradiation effect on the surface morphologies and surface properties of the substrates. The as-prepared structured surfaces show an excellent superhydrophobic property with a high static contact angle (greater than 150°). The surface properties have been investigated using Fourier transform infrared (FT-IR) spectroscopy, scanning electron microscopy (SEM), and X-ray photoelectron spectroscopy (XPS). This method provides a means to control the surface morphology. As indicated by the research results, improving the surface hydrophobicity may be an effective way of overcoming surface problems.

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1. Introduction

Superhydrophobic surfaces have been studied extensively during the last few decades because of their low cost, easy processing, low weight, high-quality surfaces, easy fabrication of thickness, etc. [1–3]. Over the past few decades, superhydrophobic surfaces have been used for a range of applications, such as microfluidics transportation, self-cleaning traffic indicators, prevention of snow adhesion on antennas and windows, reduction of frictional drag on ship hulls, and stain-resistant textiles [4–8]. The phenomenon of superhydrophobic surfaces with a water contact angle (CA) larger than 150° causes water falling on leaves to bead and roll off the surface, thereby leading to anti-corrosion, anti-icing, water repellency and self-cleaning characteristics [9–11].

Generally, artificial superhydrophobic surfaces can be achieved by controlling two different properties: either an appropriate surface morphology or a low surface energy. The combination of nanoscale hierarchical structures with low-surface-energy materials enables the fabrication of numerous artificial superhydrophobic surfaces through facile methods that utilize readily available raw materials. The methods to induce a superhydrophobic surface can be divided into three categories: a) top-down processes such as nano-imprinting, template-based techniques, and e-beam and

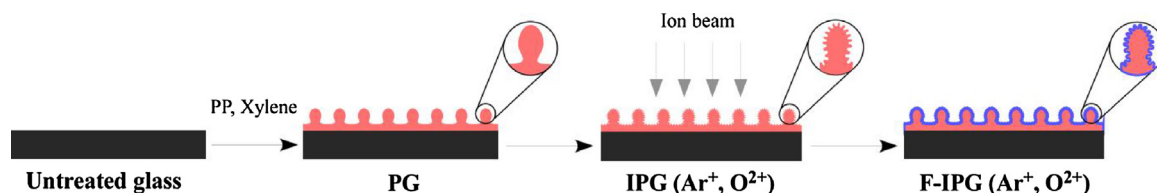
plasma treatments [12–14]; b) bottom-up processes such as layer-by-layer deposition, hydrogen bonding, and colloidal assemblies [15–17]; and c) a combination of both top-down and bottom-up processes such as the casting of polymer solutions, phase separation, and electro-spinning with chemical vapor deposition [18–20].

Among these methods, ion beam irradiation has been widely used to produce a hydrophobic surface on substrates. This process is known to be a rapid and environmentally friendly method, and it is quite controllable; the type, energy, and fluence of the ion beam may be selected to fit various applications [21–30].

In this study, we fabricated nanostructured superhydrophobic surfaces using ion beam irradiation. Under the ion irradiation of keV Ar^+ and O^{2+} ions, we demonstrated that different substrate morphologies exhibited relatively different superhydrophobic surface properties [29–31]. The nanostructured superhydrophobic surfaces were characterized by measuring the static CA. The obtained surface exhibited superhydrophobicity with a higher static contact angle (160° and 162°). Chemical structure and morphology analyses were performed, focusing on the change in the chemical structure and surface morphology of the substrates before and after ion beam irradiation. It is found that the ion-irradiated superhydrophobic surfaces present morphological structures similar to those of lotus leaves, thus presenting a reasonable and optimized process.

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Scheme 1. Schematic illustration of the fabrication procedure of the superhydrophobic surfaces.

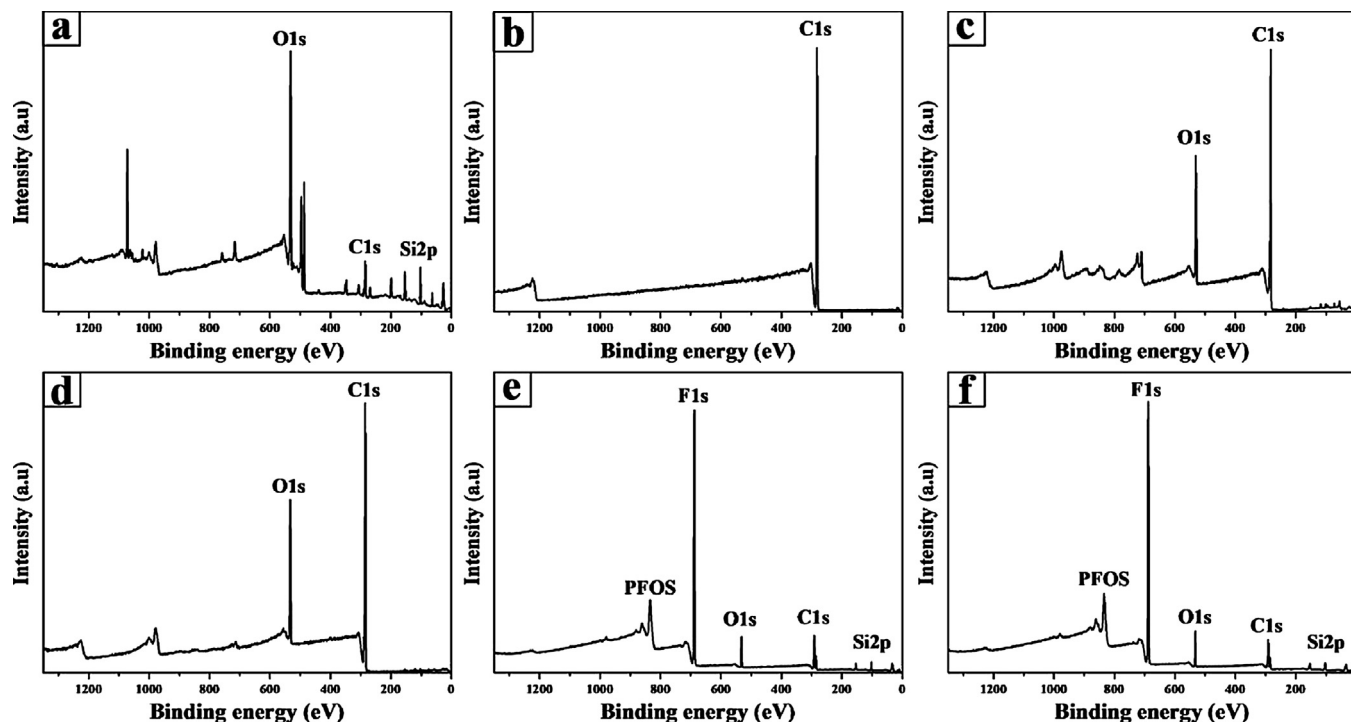


Fig. 1. XPS survey spectra of (a) Untreated glass, (b) PG, (c) IPG(Ar⁺), (d) IPG(O₂⁺), (e) F-IPG(Ar⁺), (f) F-IPG(O₂⁺).

2. Experimental

2.1. Materials

Polypropylene (PP) and trichloro(1*H*,1*H*,2*H*,2*H*-perfluorooctyl) silane (PFOS) were purchased from Sigma-Aldrich. The reagent-grade solvents, toluene, xylene, and acetone were purchased from TCI.

2.2. Preparation of superhydrophobic coating

First, 2.4 g of granulated PP was added to a volumetric flask containing 40 mL of xylene. 2.4 g was chosen as the upper bound because the use of a larger amount of PP to form a homogeneous solution was unsuccessful. The temperature of the PP solution was increased to 120 °C. The solution was dripped onto the cleaned glass substrate. The PP surface that formed was dried at a given temperature at 60° to remove the solvent. Then, the PP surface samples with dimensions of 20.0 × 20.0 mm² were placed on a rotating roll at a speed of 10 mm/s. The vacuum system was evaluated at a base pressure of 9.7×10^{-5} Torr. Discharge gases (Ar, O₂) were introduced into the ion beam source. The gas flow rates were 30 and 80 sccm, and the working pressures were approximately 0.6 and 2.7 mTorr for the Ar and O₂, respectively. A DC high-voltage power supply (3KL1000; Glassman) was used. The discharge voltage and current were 1.8 kV (Ar) and 0.8 kV (O₂) and 140 ± 10 mA, respectively. For the ion beam irradiation, the amounts of ions were 4.756×10^{16} and 5.649×10^{15} ion/cm². Finally, the synthetic process involved a uniform spreading of a few drops (30 μL) of PFOS with a Pas-

teur pipette on an ion-irradiated PP surface substrate, followed by an UV laser pulse for specific time-periodic intervals ranging from 15–30 min. Scheme 1 depicts the images of the process.

2.3. Characterization

The surface morphologies were analyzed by field-emission scanning electron microscopy (FE-SEM, JSM-7000F, JEOL, Ltd.), while X-ray photoelectron spectroscopy (XPS, K-Alpha, Thermo Fisher, Ltd.) techniques were used to obtain the chemical compositions of the modified samples. The surface wetting behaviors were assessed by the water contact angle (CA) which is collected by a CA goniometer (DSA 100, Krüss, Ltd.). Water droplets with a volume of 3 μL were carefully dropped onto the surface in ten different positions to obtain the average static CA value. The infrared spectra of the samples were recorded with a Fourier transform infrared (FT-IR, VERTEX-70, Bruker Optics, Ltd.) spectrometer.

3. Results and discussion

3.1. Surface elemental composition

The elemental compositions of the fabricated surface samples were analyzed using XPS and FT-IR spectroscopy. XPS is one of the surface-sensitive techniques used to provide information on the changes in the surface chemistry. The elemental compositions of the surfaces of the untreated glass, PG, IPG (Ar, O₂), and F-IPG series are displayed in Fig. 1. It can be seen from these spectra that the main components on the surface are fluorine (F), oxy-

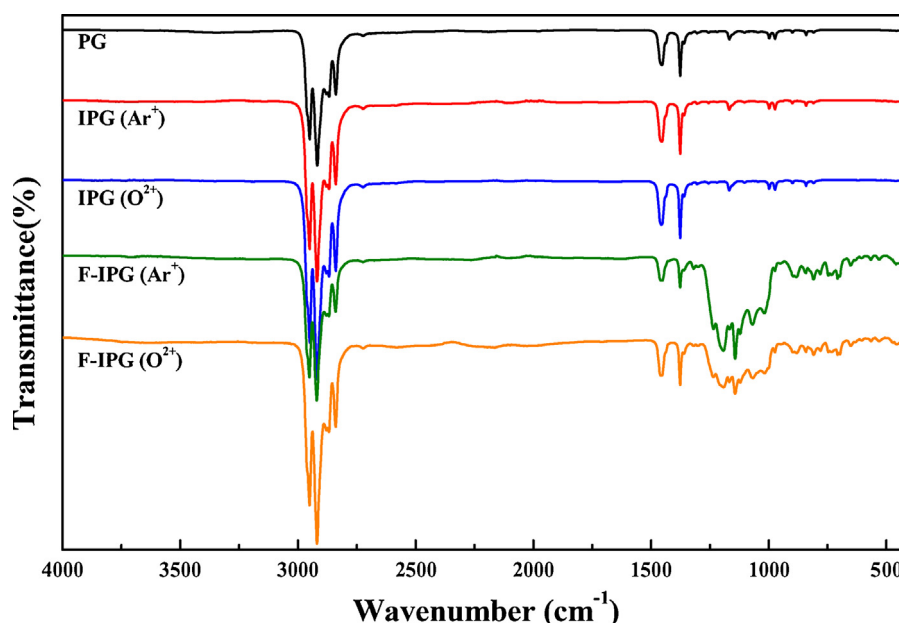


Fig. 2. FT-IR spectra of PG, IPG(Ar⁺, O₂⁺) and F-IPG(Ar⁺, O₂⁺) samples.

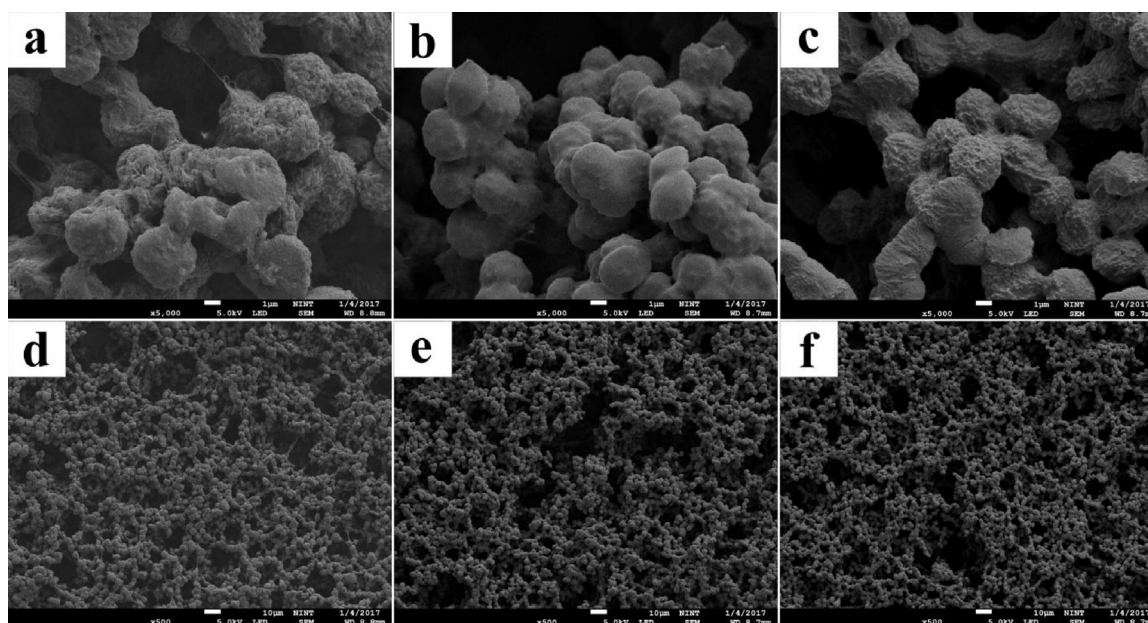


Fig. 3. SEM images showing the surface morphologies of (a) and (d) PG, (b) and (e) IPG(Ar⁺), (c) and (f) IPG(O₂⁺).

gen (O), and carbon (C). Fig. 1(b) shows a C 1s peak, which was attributed to the PP. The almost completely disappeared except for the carbon C 1s peak, indicating that PP coated almost all areas of the untreated glass. The peaks of C 1s and O 1s seen in Fig. 1(c) and (d) indicate that Ar⁺ and O₂⁺ ions were irradiated on the surfaces. The intensity of the O 1s peak in Fig. 1(c) and (d) obviously increased, which confirmed that Ar⁺ and O₂⁺ were irradiated on the PP-coated surface. In Fig. 1(e) and (f), a signal was observed at 688.36 eV in the survey spectrum of the fluorine, indicating the successful fluorination of the irradiated PP surface. In addition, the surface functional groups were determined by the FT-IR spectra displayed in Fig. 2. The appearance of absorption peaks at 1376, 1456, 2838, 2918, and 2950 cm⁻¹ for the PP-coated and irradiated (Ar, O₂) samples was observed (Fig. 2(a)–(c)). Absorption peaks are found at 2950 cm⁻¹ and 2917 cm⁻¹, corresponding to the –CH₃

group in the typical spectrum. The peak frequencies at 1454 cm⁻¹ and 2838 cm⁻¹ are due to symmetric and asymmetric stretching vibrations, respectively, of CH₂ in the alkyl chain. The existence of functional groups on the PFOS-coated surfaces is confirmed in Fig. 2(d) and (e). The peaks at 1142 cm⁻¹ and 1192 cm⁻¹ are the symmetric and asymmetric stretching frequencies, respectively, of CF₂ in the perfluoroalkyl chain. The peak found around 1068 cm⁻¹ and the peak observed at 840 cm⁻¹ are due to the Si–C bonds.

3.2. Surface morphological studies

SEM was used to observe the morphologies of the surfaces after irradiation with different ion beams. The samples were treated using Ar⁺ and O₂⁺ ion beams with different ion doses. Fig. 3 shows the images of rough surfaces before and after ion irradiation with

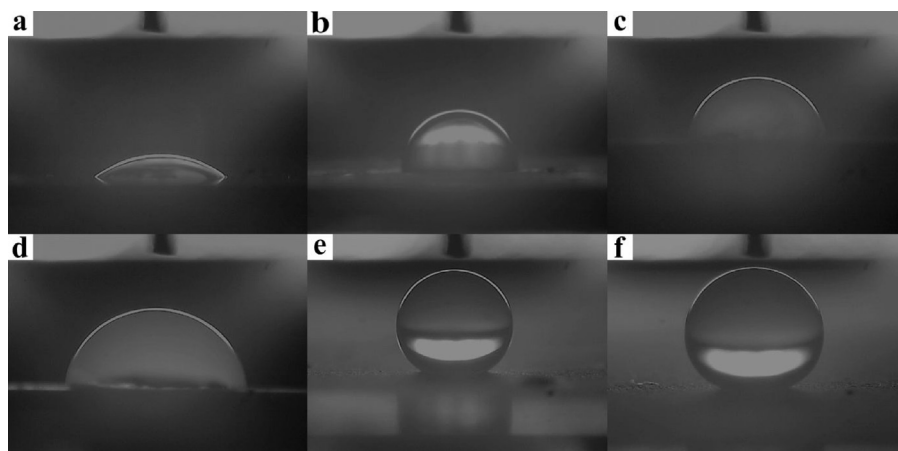


Fig. 4. Water contact angle images of (a) Untreated glass, (b) PG, (c) IPG(Ar⁺), (d) IPG(O₂⁺), (e) F-IPG(Ar⁺), (f) F-IPG(O₂⁺).

ion species (Ar, O₂) at fixed dose conditions of 4.756×10^{16} and 5.649×10^{15} ion/cm², respectively. After irradiation, the surface of the PP-coated sample became quite rough. This change can be seen more clearly in Fig. 3 with a larger magnification. In contrast to the surface of the PP-coated sample, the surfaces of the irradiated samples confirm the formation of small protuberances at the nanometer scale which are uniformly spaced at distances on the order of micrometers (Fig. 3(b) and (c)). These formed structures should be attributed to the sputtering of ions from the surface caused by Ar⁺ and O₂⁺ ion irradiation. The air trapped between the protuberances formed a layer with a low surface energy which results in superhydrophobicity. This counting result indicates that the topological structure of the irradiated surface is a key point to induce such superhydrophobic surfaces.

To describe the superhydrophobic surface, the water CAs were tested using an optical CA instrument. As shown in Fig. 4(e) and (f), the rough PFOS surface with hierarchical surface topography was successfully fabricated, demonstrating superhydrophobicity (static CA is $160^\circ \pm 5^\circ$). It was observed that the CA changed along with the roughness, which was affected by the low-surface-energy materials. In comparing the prepared rough structures to the untreated surfaces, (Fig. 4(a)) it can be seen that the roughness of the PP-coated surfaces increased (Fig. 4(b)), but that of the ion (Ar⁺, O₂⁺) irradiated surfaces decreased (Fig. 4(c) and (d)), and the surfaces modified by PFOS had the largest CAs of 160° and 162° (Fig. 4(e) and (f)), respectively. This result is the average of 10 data points acquired from different areas of the surface, while the error shows the deviation. The hysteresis of a liquid on a solid surface is as important as the static CA for a superhydrophobic surface when considering the self-cleaning effect. This situation can be modeled by the Cassie–Baxter model, which describes the wetting behavior of a surface through the following equations:

$$\cos\theta_m = f_1 \cos\theta - f_2 \quad (1)$$

$$f_1 + f_2 = 1 \quad (2)$$

where θ_m and θ correspond to the static CA of the water droplets on the rough surface and untreated surface, respectively, while f_1 and f_2 represent the area fractions of PFOS and air on the surface, respectively. In the case of the fabricated superhydrophobic surface, θ_m and θ are about 162° and 35° , respectively. By using Eqs. (1) and (2), f_1 and f_2 can be correspondingly calculated as 0.027 and 0.973. These findings indicate that only 2.7% of the water droplet on the present superhydrophobic surface is in contact with the PFOS, while the remaining 97.3% of the water droplet is suspended over the 'air' area. As shown in Fig. 4, the fabricated surfaces exhibit

superhydrophobic characteristics, demonstrating the practicality of the fabricated surface.

4. Conclusion

We have presented a simple method for producing a superhydrophobic surface based on an ion (Ar⁺, O₂⁺) irradiation technique. A technologically viable substrate process for producing superhydrophobic surfaces has been developed. Untreated glass, PG (PP coated glass), IPG (irradiated PG by Ar⁺ and O₂⁺ ions), and F-IPG (PFOS coated IPG) samples were structured using this process which created a fine nanoscale surface roughness superimposed on larger microscale surface protrusions. This resulted in an increase in the CA from 35° to 105° for PG; 105° to 70° and 75° for IPG (Ar⁺, O₂⁺), respectively; and 70° and 75° to 160° and 162° for F-IPG (Ar⁺, O₂⁺), respectively. Physical and chemical structural changes of the F-IPG surface caused by ion (Ar⁺, O₂⁺) irradiation were investigated by SEM and XPS, respectively. Protuberances at the nanometer scale and spacings at distances on the order of micrometers were formed after irradiation. This formed structure is considered to be the key point to the generation of such a superhydrophobic surface. In contrast to the topological change of the surface, the formation of an irradiation effect of the surface is inferred to have a negative contribution to the hydrophobicity of the IPG surface by comparing our results with other works. However, the topological element has taken a dominant place which results in the ultimately superhydrophobic optimization of the surface. Based on the experimental results, it is expected that the production of a superhydrophobic surface would be widely applicable for self-cleaning, such as clothing and glass construction, as well as other uses.

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References

- [1] H.Y. Erbil, A.L. Demirel, Y. Avci, O. Mert, Transformation of a simple plastic into a superhydrophobic surface, *Science* 299 (2003) 1377–1380.
- [2] T. Xhu, C. Cai, C. Duan, S. Zhai, S. Liang, Y. Jin, N. Zhao, J. Xu, Robust polypropylene fabrics super-repelling various liquids: a simple, rapid and scalable fabrication method by solvent swelling, *ACS Appl. Mater. Interfaces* 7 (2015) 13996–14003.
- [3] X. Zhang, N. Zhao, S. Liang, X. Lu, X. Li, Q. Xie, X. Zhang, J. Xu, Facile creation of biomimetic systems at the interface and in bulk, *Adv. Mater.* 20 (2008) 2938–2946.

- [4] M. He, J. Wang, H. Li, X. Jin, J. Wang, B. Wang, B. Liu, Y. Song, Super-hydrophobic film retards frost formation, *Soft Matter*. 6 (2010) 2396–2399.
- [5] B. Bhushan, Y.C. Jung, K. Koch, Self-cleaning efficiency of artificial superhydrophobic surfaces, *Langmuir* 25 (2005) 3240–3248.
- [6] E. Yu, S.C. Kim, H.J. Lee, K.H. Oh, M.W. Moon, Extreme wettability of nanostructured glass fabricated by non-lithographic anisotropic etching, *Sci. Rep.* 5 (2015) 9362–9367.
- [7] Y. Liu, X. Li, J. Jin, J. Liu, Y. Yan, Z. Han, L. Ren, Anti-icing property of bio-inspired micro-structure superhydrophobic surfaces and heat transfer model, *Appl. Surf. Sci.* 400 (2017) 498–505.
- [8] S. Pan, N. Wang, D. Xiong, Y. Deng, Y. Shi, Fabrication of superhydrophobic coating via spraying method and its applications in anti-icing and anti-corrosion, *Appl. Surf. Sci.* 389 (2016) 547–553.
- [9] X. Deng, L. Mammen, H.J. Butt, D. Vollmer, Candle soot as a template for a transparent robust superamphiphobic coating, *Science* 335 (2012) 67–70.
- [10] T. Verho, C. Bower, P. Andrew, S. Franssila, O. Ikkala, R.H.A. Ras, Mechanically durable superhydrophobic surfaces, *Adv. Mater.* 23 (2011) 673–678.
- [11] X. Hao, L. Wang, D. Lv, Q. Wang, L. Li, N. He, B. Lu, Fabrication of hierarchical structures for stable superhydrophobicity on metallic planar and cylindrical inner surfaces, *Appl. Surf. Sci.* 325 (2015) 151–159.
- [12] P.P. Peng, Q. Ke, G. Zhou, T. Tang, Fabrication of microcavity-array superhydrophobic surfaces using an improved template method, *J. Colloid Interface Sci.* 395 (2013) 326–328.
- [13] R.A. Singh, E.S. Yoon, H.J. Kim, J.S. Kim, H.E. Jeong, K.Y. Suh, Replication of surfaces of natural leaves for enhanced micro-scale tribological property, *Mater. Sci. Eng. C* 27 (2007) 875–879.
- [14] C. Sun, L.Q. Ge, Z.Z. Gu, Fabrication of super-hydrophobic film with dual-size roughness by silica sphere assembly, *Thin Solid Films* 515 (2007) 4686–4690.
- [15] R. Gao, J. Wang, X. Zhang, H. Yan, W. Yang, Q. Liu, M. Zhang, L. Liu, K. Takahashi, Fabrication of superhydrophobic magnesium alloy through the oxidation of hydrogen peroxide, *Colloids Surf. A* 436 (2013) 906–911.
- [16] Q. Sun, Y. Lu, J. Li, J. Cao, Self-assembly of a superhydrophobic ZnO nanorod arrays film on wood surface using a hydrothermal method, *Key Eng. Mater.* 609 (2014) 468–471.
- [17] B.S. Ming, J.L. Li, F. Kang, G.Y. Pang, Y.K. Zhang, L. Chen, J.Y. Xu, X.D. Wang, Microwave-hydrothermal synthesis of birnessite-type MnO₂ nanospheres as supercapacitor electrode materials, *J. Power Sources* 198 (2012) 428–431.
- [18] J. Lin, Y. Cai, X. Wang, B. Ding, J. Yu, M. Wang, Fabrication of biomimetic superhydrophobic surfaces inspired by lotus leaf and silver ragwort leaf, *Nanoscale* 3 (2011) 1258–1262.
- [19] L. Camilli, M. Scarselli, S.D. Gobbo, P. Castrucci, F. Nanni, E. Gautron, S. Lefrant, M.D. Crescenzi, The synthesis and characterization of carbon nanotubes grown by chemical vapor deposition using a stainless steel catalyst, *Carbon* 49 (2011) 3307–3315.
- [20] J.H. Lee, S.H. Lee, D.Y. Kim, Y.S. Park, The structural and surface properties of carbon nanotube synthesized by microwave plasma chemical vapor deposition method for superhydrophobic coating, *Thin Solid Films* 546 (2013) 94–97.
- [21] Y. Chen, Z. Zhao, Y. Liu, Wettability characteristic of PTFE and glass surface irradiated by keV ions, *Appl. Surf. Sci.* 254 (2008) 5497–5500.
- [22] Y. Chen, Z. Zhao, J. Dai, Y. Liu, Topological and chemical investigation on super-hydrophobicity of PTFE surface caused by ion irradiation, *Appl. Surf. Sci.* 254 (2007) 464–467.
- [23] S.H. Lee, E.Y. Byun, J.K. Kim, D.G. Kim, Ar and O₂ linear ion beam PET treatments using an anode layer ion source, *Curr. Appl. Phys.* 14 (2014) S180–S182.
- [24] Y. Yamada, K. Takahashi, T. Ikuta, T. Nishiyama, Y. Takata, W. Ma, A. Takahara, Tuning surface wettability at the submicron-scale: effect of focused ion beam irradiation on a self-assembled monolayer, *J. Phys. Chem. C* 120 (2016) 274–280.
- [25] J.H. Oh, T.J. Ko, M.W. Moon, C.H. Park, Nanostructured superhydrophobic silk fabric fabricated using the ion beam method, *RSC Adv.* 4 (2014) 38966–38973.
- [26] G.M. Shanthini, C.A. Martin, N. Sakthivel, S.C. Veerla, K. Elayaraja, B.S. Lakshmi, K. Asokan, D. Kanjilal, S.N. Kalkura, Physical and biological properties of the ion beam irradiated PMMA-based composite films, *Appl. Surf. Sci.* 329 (2015) 116–126.
- [27] W.Y. Ding, D.Y. Ju, W.P. Chai, The effect of working pressure on the chemical bond structure and hydrophobic properties of PET surface treated by N ion beams bombardment, *Appl. Surf. Sci.* 256 (2010) 6876–6880.
- [28] A. Metya, D. Ghose, N.R. Ray, Development of hydrophobicity of mica surfaces by ion beam sputtering, *Appl. Surf. Sci.* 293 (2014) 18–23.
- [29] A.M.A. Reheem, M.I.A.A. Maksoud, A.H. Ashour, Surface modification and metallization of polycarbonate using low energy ion beam, *Radiat. Phys. Chem.* 125 (2016) 171–175.
- [30] E. Francisco, Isaac C. Sanchez, Beatriz L. España-Sánchez, M. Josue D. C. Salvador, P. Fabienne, L. Gabriel, Scanning-probe-microscopy of polyethylene terephthalate surface treatment by argon ion beam, *Nucl. Instrum. Method B* 362 (2015) 49–56.
- [31] S.W. Lee, J.W. Hong, M.Y. Wye, J.H. Kim, H.J. Kang, Y.S. Lee, Surface modification and adhesion improvement of PTFE film by ion beam irradiation, *Nucl. Instrum. Method B* 219 (2004) 963–967.